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ITAI 2376 - Deep Learning

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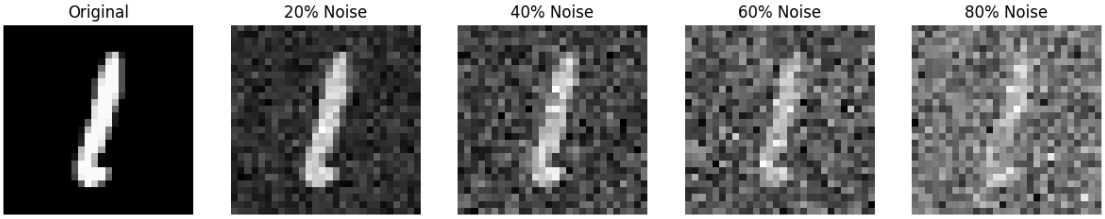
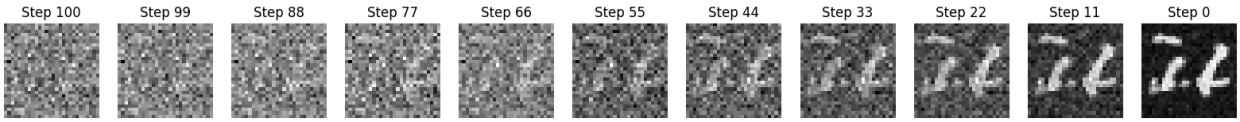
L08 - Diffusion Models

1. Understanding Diffusion

- **Explain what happens during the forward diffusion process, using your own words and referencing the visualization examples from your notebook.**
    - During forward diffusion, the model starts with a real image and gradually adds small amounts of random Gaussian noise until the image becomes almost pure noise. In the notebook visualization, the digit started clear and then became blurrier and more distorted at each step until it was hard to recognize. This matches the module 08 powerpoint which describes forward diffusion process as slowly 'destroying an image'.



- **Why do we add noise gradually instead of all at once? How does this affect the learning process?**
    - We add noise gradually instead of all at once to make learning easier. If the image were destroyed in one jump then the model would have to recover everything at once from pure randomness. Adding noise in small steps makes it so that the U-Net only has to predict and remove a limited amount of noise at each timestamp. This makes training more stable and manageable. The training loop in my notes is built around this idea of picking an image, picking a timestamp, adding that amount of noise, and then asking the U-Net to predict the noise that was added.
- **Look at the step-by-step visualization - at what point (approximately what percentage through the denoising process) can you first recognize the image? Does this vary by image?**
    - Based on the denoising progression images, there is definitely a lot of noise for most of the steps. It becomes somewhat recognizable around the last quarter of the process after Step 22 and clearer on Step 0. However even though the image does get clear, it doesn't really show a strong or correct handwritten digit.

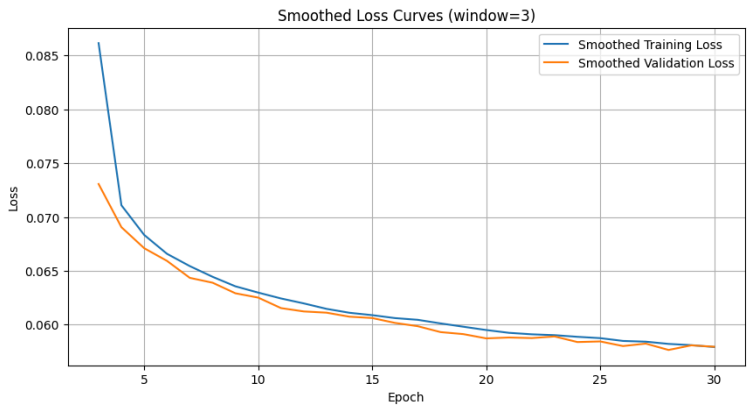


2. Model Architecture

- **Why is the U-Net architecture particularly well-suited for diffusion models? What advantages does it provide over simpler architectures?**
    - The U-Net architecture is well suited for diffusion models because it is the main denoiser that predicts the noise to remove at each step. In this notebook, the U-Net has a downsampling path, middle blocks, and an upsampling path that outputs predicted noise and the same image shape as the input. This helps it learn the overall digit structure and finer image details while predicting the noise to remove.
- **What are skip connections and why are they important? Explain them in relations to our model**
    - Skip connections are shortcuts that send information from the earlier of the model to the later part. In this model, they help keep important image details from getting lost while the image is being shrunk and rebuilt. This is useful because the model is trying to remove noise without losing the shape of the digit. Module 08 says that skip connections are important because diffusion models need precise, pixel-level predictions.
- **Describe in detail how our model is conditioned to generate specific images. How does the class conditioning mechanism work?**
    - This model is conditioned using both the digital label and the timestep. In the notebook, the digital label is turned into a one-hot vector and then passed through the class embedding block while the timestep goes through the time embedding block. These two pieces of information are added into the middle of the U-Net so the model knows what number it is supposed to generate and how much noise is in the image at that step. The notes explain that this is what makes conditional diffusion useful. Without conditioning the model it would just generate random images instead of a chosen class.

3. Training Analysis (20 points)

- **What does the loss value tell of your model tell us?**
    - The loss tells us how well the model is predicting the noise that was added to each image. In diffusion training, the U-Net's main job is to predict the noise , and the notebook compares predicted noise to actual noise using MSE loss. In my results, both training and validation loss went down a lot (from about 0.11 and 0.08 to about 0.058) which shows the model did learn the noise-prediction task. Since the two losses stayed very close, the training looked stable and did not show strong overfitting.

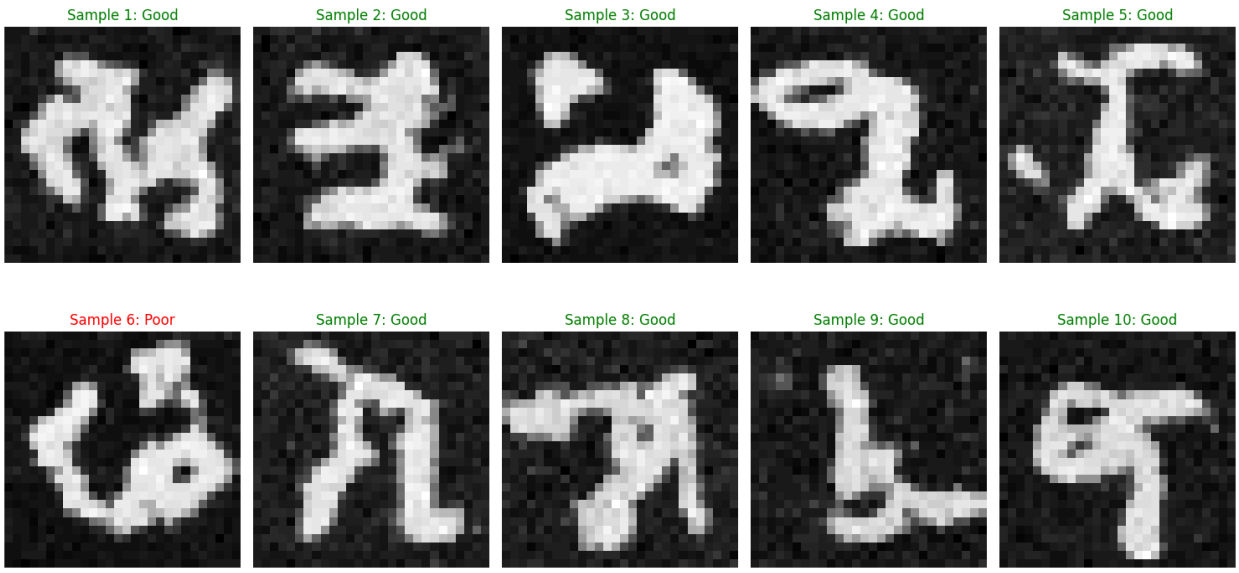


- **How did the quality of your generated images change change throughout the training process?**
    - Honestly speaking, even though the loss improved the final images were still weak and didn't really look like clear handwritten digits. So the training helped the model learn the task better but that did not fully turn into strong visual quality. I had done the notebook once before but was confused as to why the images turned out like that. I decided to redo it slowly step by step to make sure the weak digits were not caused by a coding mistake.

- **Why do we need the time embedding in diffusion models? How does it help the model understand where it is in the denoising process?**
    - We need time embeddings because the model has to know how much noise is in the image at the current step. Early in the reverse process the image is mostly noise but changes each step. The module 08 booklet mentions sinusoidal encoding which turns timestep numbers into vectors which are then added into the U-Net features. Time embedding basically tells the model where it is in the denoising process so it knows how much cleanup to do.

4. CLIP Evaluation (20 points)

- **What do the CLIP scores tell you about your generated images? Which images got the highest and lowest quality scores?**
    - Based on what the notebook says, CLIP scores are used here to evaluate how realistic the images that are being generated are. In my results, CLIP recognized 90% of the digit 6 samples as good examples. At the same time, the images still looked weak to me so the scores seem more positive than the visual result. Since there wasn't a clear score for the results, I asked ChatGPT for help to write a code to show them. Here I was able to see that Sample 9 had the highest quality score because the 'clear' score was the highest at 93.5% and a low 'blurry' score of 4.7%. The weakest example was Sample 6 where the 'clear score' was 45.5% and the 'blurry' score was 52.4%. I thought this was interesting because sample 6 looked more like a 6 to me than the others.



- **Develop a hypothesis explaining why certain images might be easier or harder for the model to generate convincingly.**
    - I think some digits are easier because they have simpler overall shapes while others can be easy to confuse with similar looking numbers. If the model learns only rough structure and not fine detail it can end up looking messy. Also CLIP may still rate an image fairly good even if the image doesn't look very convincing to a person. That would explain why the CLIP score looked strong for digit 6 even though the images were still distorted.
- **How could CLIP scores be used to improve the diffusion model's generation process? Propose a specific technique.**
    - CLIP scores can be used as a ranking step after generation. The model could generate several versions of the same digit and then with the CLIP scores, keeping only the samples with the best match. This would improve the final outputs.

5. Practical Applications (20 points)

- **How could this type of model be useful in the real world?**
    - This type of model is useful because diffusion can generate new images from noise in a controlled way. Module 08 mentions applications like text to image generation and image editing.
- **What are the limitations of our current model?**
    - The biggest limitation is that the images are still not very clear even though the loss values improved. Another limitation is that the model is slow because diffusion generates images step by step instead of in one pass. This version is also very small and trained only on MNIST so it is good for learning but not strong enough for realistic image generation.
- **If you were to continue developing this project, what three specific improvements would you make and why?**

- To address the mismatch between numerical success and visual quality I would use the technique mentioned in one of the earlier questions. I would generate multiple samples, use CLIP to score them and then keep the best one while also using human judgement to make sure. Since module 08 explains that conditional diffusion is what makes the output controllable, I would also focus on improving that part. I would also try a faster and better sampler since improving the sampling stage is one of the most direct ways to try to get better outputs